Leam Howe - Curriculum Vitae

PhD Researcher, University of Edinburgh, SENSE Earth Observation CDT

Research Interests

PhD researcher in Environmental science with a particular focus on remote sensing and Earth observation from space; keen on utilising machine learning to capitalise on these rapidly expanding datasets. Developing a career in academia whilst maintaining relationships with industry partners and working on outreach projects to improve public communication of climate science. Partial to mountains, snow, and coffee.

Programming & Software: Python, git, shell, ssh, Slurm, Matlab, Fortran, Javascript, html, LaTeX, QGIS, ArcGIS, Google Earth Engine, Planetary Computer, Jupyter Notebooks, Docker, AWS, Google Colab, Hugging Face, PyTorch, Scikit-learn.

Relevant Technical Expertise: Multispectral Optical Imagery (Landsat, Sentinel-2, Planet) | SAR Imagery (Sentinel-1) | Digital Elevation Models | Meteorological Reanalysis | Snow Modelling | Machine Learning | UAV Remote Sensing | Fieldwork in Mountainous Environments

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Education

2022 - Present: University of Edinburgh, PhD NERC SENSE Earth Observation CDT

Remote sensing and modelling of mountain snow patches. Using novel high-resolution remote sensing and meteorological modelling to improve our understanding of the climate sensitivity of mountain snow.

2021 – 2022: University of Leeds, MRes Climate and Atmospheric Science (Distinction) Research project applying machine learning techniques to efficiently identify anomalous behaviour in SAR ice velocity data.

2015 – 2019: University of Nottingham, MSci Physics (2:1 Hons) Two first-class research projects and gained valuable coding, statistical analysis, and communication skills.

Publications in review

Howe, L., Essery, R., Crowley E. J., SPUN: Deep learning for continuous snow cover fraction retrieval in marginal environments. *in review with Hydrology and Earth System Sciences (HESS)*

Non Peer-reviewed Publications

Howe, L. (2026) The Death of Scotland's Zombie Glaciers. *Published in RSGS's 'The Geographer'.*

Grants

- **Global Change Institute small grants - £850** (2025)

Funded to set up a new 'Machine Learning for Geoscience and Earth Observation (ML4GEO)' research group

- **NERC Diversifying the Pipline** (2024)

Jointly awarded to the SatSchool outreach project and the SENSE CDT and E4 DTP to support the development of outreach materials to engage underrepresented groups in STEM.

- **SENSE outreach funding - £10,000**

Awarded to the SatSchool outreach project from the SENSE CDT to contribute to continued development, refinement, and delivery of outreach materials.

- **SAGES Small Grant Scheme for Scottish fieldwork - £500** (2024)

Fieldwork in the Cairngorms, Scotland, contributing to my PhD project investingating the reflective properties of Scottish snow using a sensor matched to Sentinel-2 and mounted on the DJI M350 drone.

- **SAGES Small Grant Scheme - £850**

Awarded to run a workshop to promote engagement with the SatSchool outreach project and developed new ideas for outreach materials.

- **Ogden Trust Physics Education Grant - £5,000** (2022)

Grant awarded to SatSchool Outreach projec to align outreach resources with the KS3 Physics curriculum.

- **SENSE Earth Observation CDT Studentship - £66,000** (2022 - Present)

PhD fully funded by the UKRI Natural Environment Research Council.

Awards & Prizes

- **SAGES ASM 2026 Best talk prize**, awarded for an oral presentation on my PhD research. ~ May 2026
- **SENSE Student Award for Outreach** ~ April 2026
- **SENSE Industry Symposium prize for soical contribution**, awarded for efforts to develop and promote a collaborative and social community within the SENSE CDT. ~ November 2023

Oral Presentations

- **SAGES ASM 2026, May 2026** From Scottish Snow Patches to Global Snow Cover: Deep Learning for Snow Cover Fraction Retrieval
- **SAGES AI/ESM Fora Meeting, October 2025** Invited speaker.
- **Living Planet Symposium, Vienna July 2025**
- **SLF External Seminar, September 2024** Invited speaker
- **National Earth Observation Conference, York Sep 2024**
- **SAGES Annual Science Meeting, May 2024**

Poster Presentations

- **University of Cambridge ICCS NERC workshop on Bayesian Machine Learning as a tool for Climate Scientists, April 2024**
- **8th General Assembly of the International Union of Geodesy and Geophysics (IUGG), July 11-20, 2023 in Berlin, Germany.**
- **Scottish Alliance for Geoscience & Society (SAGES) ASM 2023.**

Presentation Contributions

Essery, R., Nemec, J., **Howe, L.**, Schwaizer, G., and Nagler, T.: Gap filling satellite snow cover for mountain catchments, EGU General Assembly 2025, Vienna, Austria, 27 Apr–2 May 2025, EGU25-6832, https://doi.org/10.5194/egusphere-egu25-6832, 2025.

Fieldwork Experience

During my PhD, I have completed fieldwork in the Cairngorms, Scotland, to collect spectal reflective measurements and perform GNSS mapping of late-lying snow patches. My first field season (2023) we used the handheld ASD Field Spectrometer. The second season I qualified as a drone pilot and used the DJI M350 drone with a MAIA multispectral sensor matched to Sentinel-2.

I have also assisted the Airborne Research and Innovation (ARI) team at the University of Edinburgh with fieldwork as part of the Forest and Peatlands project aimed to restore peatlands and expand forests in Scotland.

Teaching Experience

2025 - 2026 - Undergraduate Course (SCQF Level 8): Environmental Geography (GEGR08013), including fieldwork in the Cairngorms, Scotland.

2024 - 2025 - Postgraduate Course (SCQF Level 11): Principles and Practice of Remote Sensing (PGGE11233). - Undergraduate Course (SCQF Level 9): Key Methods in Physical Geography (GEGR09018).

2023 - 2024 - Undergraduate Course (SCQF Level 8): Meteorology: Weather and Climate (METE08002)

2022 - 2023 - Postgraduate Course (SCQF Level 11): Principles and Practice of Remote Sensing (PGGE11233). - Undergraduate Course (SCQF Level 10): Ice and Climate (GEGR10119) - Supervised a Research Experience Placement student, analysing the consistency of satellite records with snow observations in Scotland.

Outreach & Volunteering

- **SatSchool Outreach:** SatSchool Committee Member (November 2022 - Present) https://satschool-outreach.github.io/. This is a valuable outreach project developing and delivering learning resources to engage lower secondary school pupils with Earth Observation. I led the project as Chair for 2024-2025, and notably our resources were used for the Committee on Earth Observation Satellites (CEOS) Youth Hub outreach international outreach initiative during this period.
- **ML4GEO:** Key founder and co-organiser of the Machine Learning for Geoscience and Earth Observation (ML4GEO) research group at the University of Edinburgh. This is a new initiative to promote interdisciplinary collaborations between machine learning researchers and geoscientists. We have organised regular meetings, workshops, seminars, and a Geospatial Foundation Model Hackathon to foster this community.

Additional Relevant Experience

ENVEO IT GmbH (October – December 2023) ENVEO is an IT company developing satellite data product of the cryosphere. I completed a 3-month research placement working with their snow products. This gave me valuable experience of coding protocols in industry whilst working with geospatial data and an insight into the business model of an academic spin-off companies.

Astrium UK (October 2013): EADS aerospace manufacturer (notably developed Cryosat and Sentinel satellites). Shadowed engineers developing satellites and spacecraft.

Other Relevant Qualifications

- Qualified drone pilot, A2 CofC
- Mountain Leader (ML) qualified, working towards Winter ML (completed training, assessment next season).

Hobbies & Interests

Outdoors/Sports: Lover of the outdoors. Often in the mountains hiking, climbing, running, or cycling. I have achieved my Advanced Open Water Diver and Underwater Digital Photographer PADI qualifications.

Art (photography, sketching, painting): I have a passion for photography and rarely travel anywhere without my camera. I enjoy sketching and watercolour, often in inspiring places.

Music: I have achieved grade 7 piano and grade 6 guitar.

References

Prof Cathryn Birch, Professor, School of Earth and Environment, University of Leeds, C.E.Birch@leeds.ac.uk, +44(0)113 343 9831

Prof Anna Hogg, Professor, School of Earth and Environment, University of Leeds, A.E.Hogg@leeds.ac.uk, +44(0)113 343 5842.

Prof Richard Essery, Professor, School of Geosciences, University of Edinburgh, richard.essery@ed.ac.uk, +44(0)131 651 9093.

Dr Steven Bamford, Associate Professor, School of Physics & Astronomy, University of Nottingham, steven.bamford@nottingham.ac.uk. +44 (0)1157 484054.